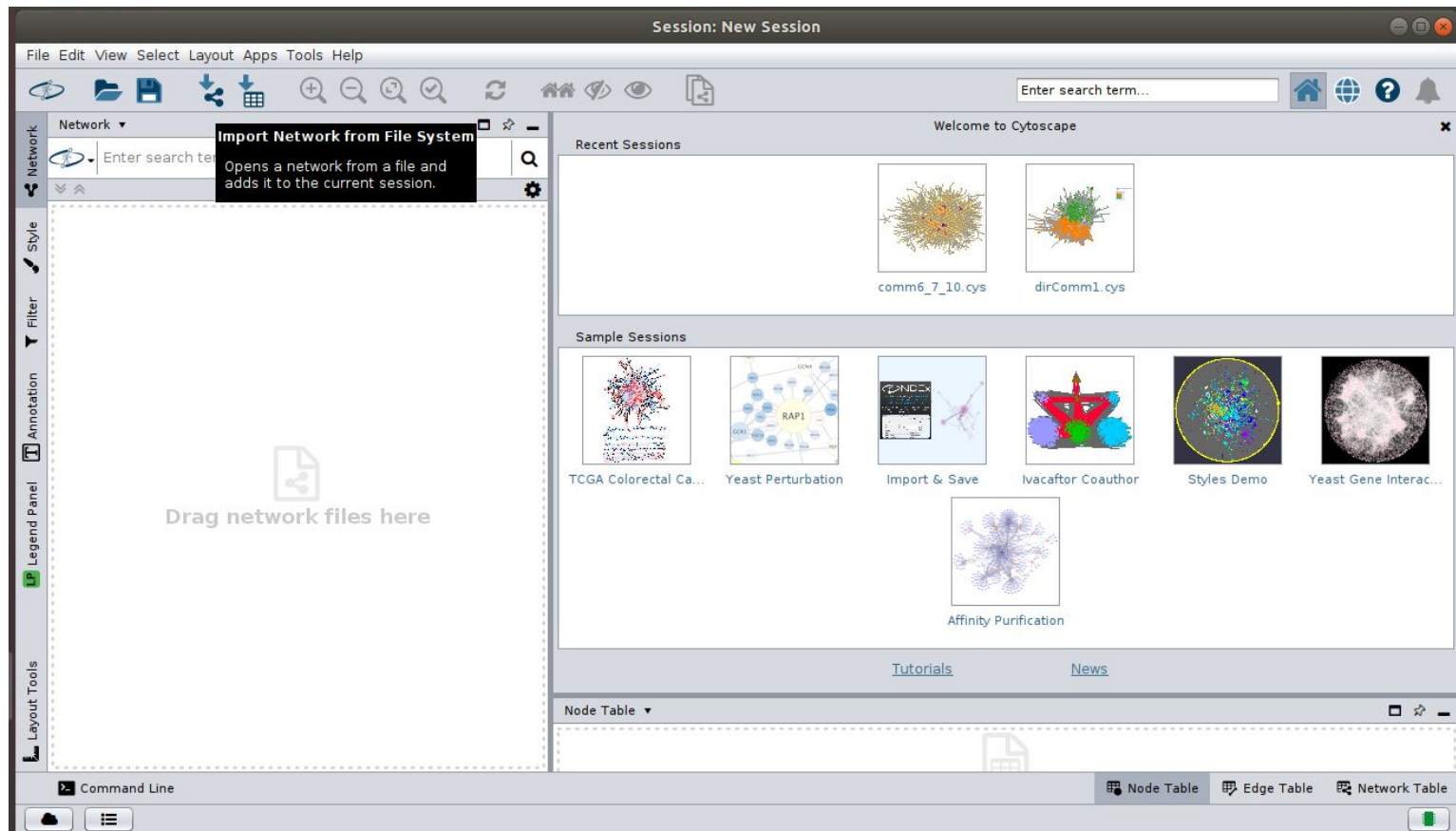


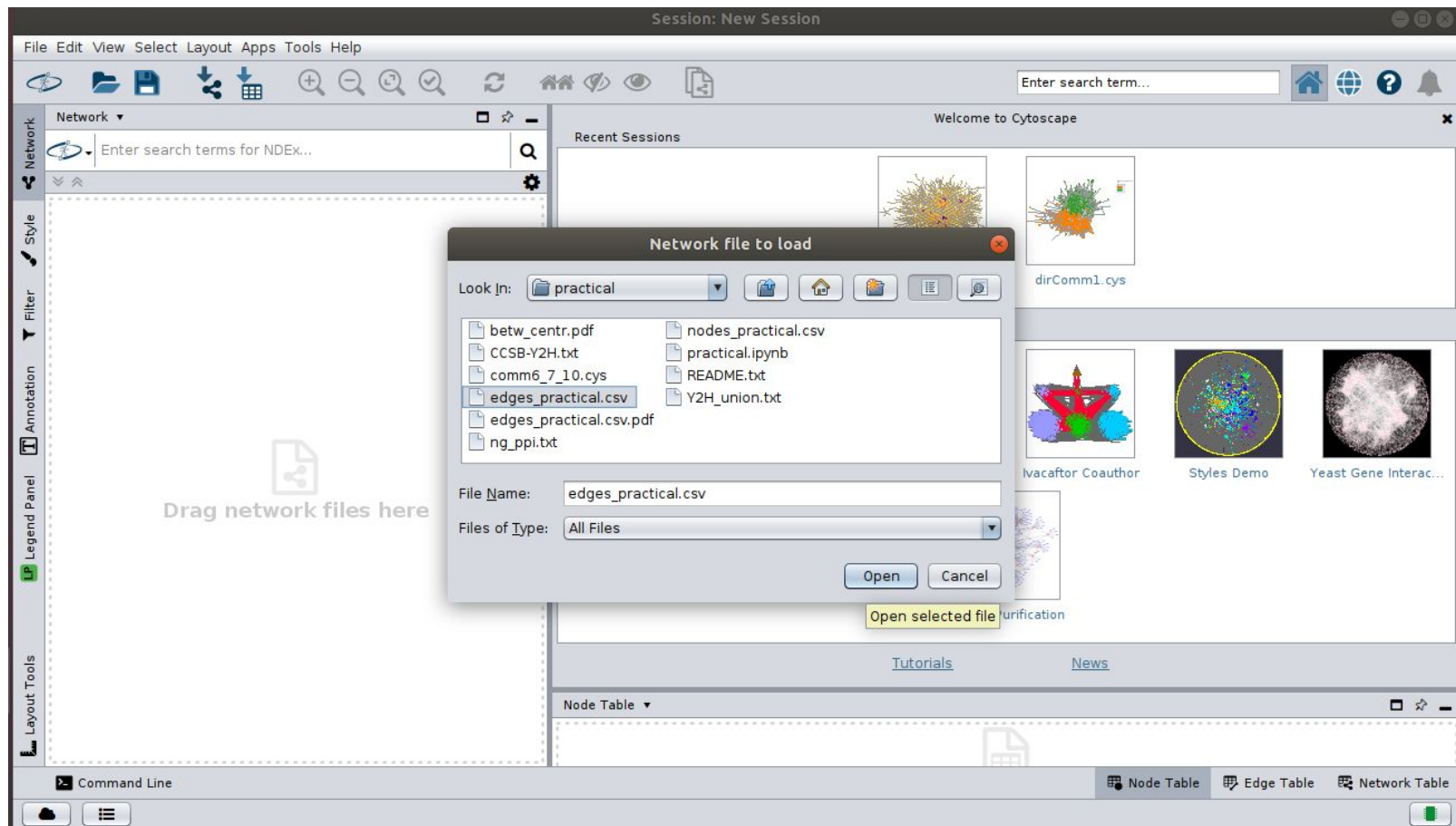
Cytoscape demo



Go to “**Import Network**” to import edge data for visualization



Select file containing your edgelist from the directory.



Session: New Session

File Edit View Select Layout Apps Tools Help

Enter search term...

Welcome to Cytoscape

Recent Sessions

Import Network From Table

Preview

Click on a column to edit it.

Select All Select None

Source	Target
P25788	O00231
P25788	O00232
P25788	O00487
P25788	O14640
P25788	O14818
P25788	O43242
P25788	O43364
P25788	O43592

Advanced Options...

OK Cancel

Styles Demo Yeast Gene Interac...

Tutorials News

Node Table

Command Line

Node Table Edge Table Network Table

Change the “**Layout**” according to the one best suited for your visualization.

The screenshot displays the Cytoscape software interface during a 'New Session'. The 'Layout' menu is open, showing various layout algorithms. The 'Compound Spring Embedder (CoSE)' option is highlighted, indicating it is the selected layout. The main canvas shows a large, complex network graph with numerous blue square nodes and black edges. A search bar at the top right contains the text 'Enter search term...'. The bottom of the interface includes a 'Table' dropdown menu with options for 'Node Table', 'Edge Table', and 'Network Table'. The status bar at the very bottom shows a 'Command Line' tab and a green indicator light.

Session: New Session

File Edit View Select Layout Apps Tools Help

Network

Style

Filter

Annotation

Legend Panel

Layout Tools

edges_practical

edges_prac

Bundle Edges

Clear All Edge Bends

Layout Tools

Settings...

Apply Preferred Layout F5

Copycat Layout

Grid Layout

Hierarchical Layout

Circular Layout

Stacked Node Layout

Attribute Circle Layout

Attribute Grid Layout

Degree Sorted Circle Layout

Prefuse Force Directed Layout

Group Attributes Layout

Edge-weighted Force directed (BioLayout)

Edge-weighted Spring Embedded Layout

Compound Spring Embedder (CoSE)

Inverted Self-Organizing Map Layout

Install yFiles Circular Layout

Install yFiles Hierarchic Layout

Install yFiles Hierarchic Layout Selected Nodes

Install yFiles Organic Layout

Install yFiles Orthogonal Layout

Install yFiles Radial Layout

Install yFiles Tree Layout

Install yFiles Orthogonal Edge Router

Install yFiles Organic Edge Router

edges_practical.csv

Table

Node Table Edge Table Network Table

Command Line

Select your network representation “**Style**” according to your preference

Session: New Session

File Edit View Select Layout Apps Tools Help

Enter search term...

Network Style Filter Annotation Legend Panel Layout Tools

Source Target

default default black

Directed Gradient1

Marquee Minimal

Nested Network Style Ripple

Sample1 Sample2

Sample3 size_rank

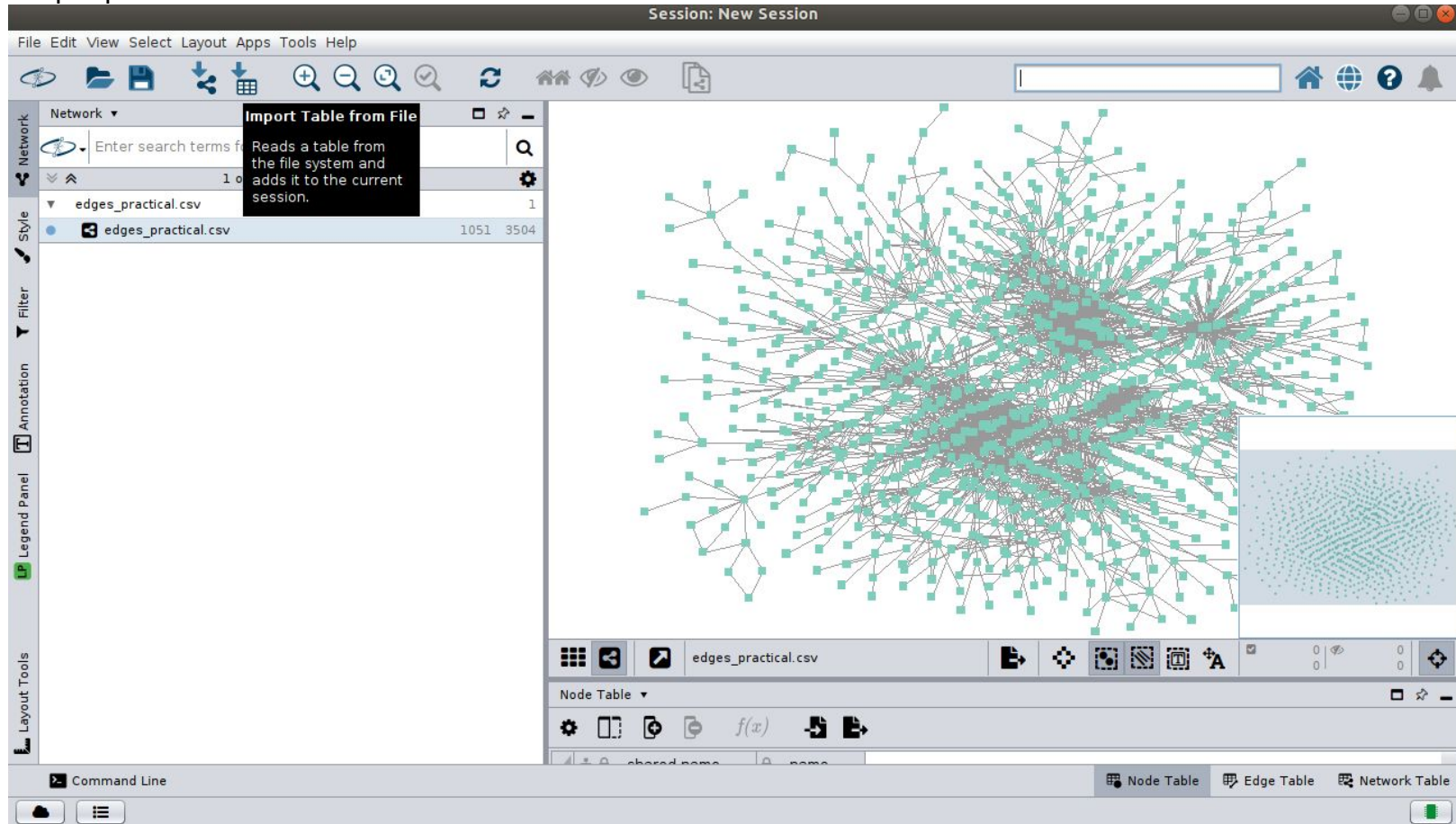
Solid Universe

edges_practical.csv

Node Table

Node Table Edge Table Network Table

Import nodes file containing node attributes for visualizing certain node properties.



Make sure that “Import data as:” should be **Node Table Columns**.

Session: New Session

File Edit View Select Layout Apps Tools Help

Network ▾ Enter search terms for NDEx... 1 of 1 Network s

edges_practical.csv

edges_practical.csv

Style

Filter

Annotation

Legend Panel

Layout Tools

Import Columns From Table

Target Table Data

Where to Import Table Data: To a Network Collection ▾

Select a Network Collection

Network Collection: edges_practical.csv ▾

Import Data as: Node Table Columns ▾

Key Column for Network: shared name ▾

Case Sensitive Key Values: ☒

Preview

Click on a column to edit it. Select All Select None

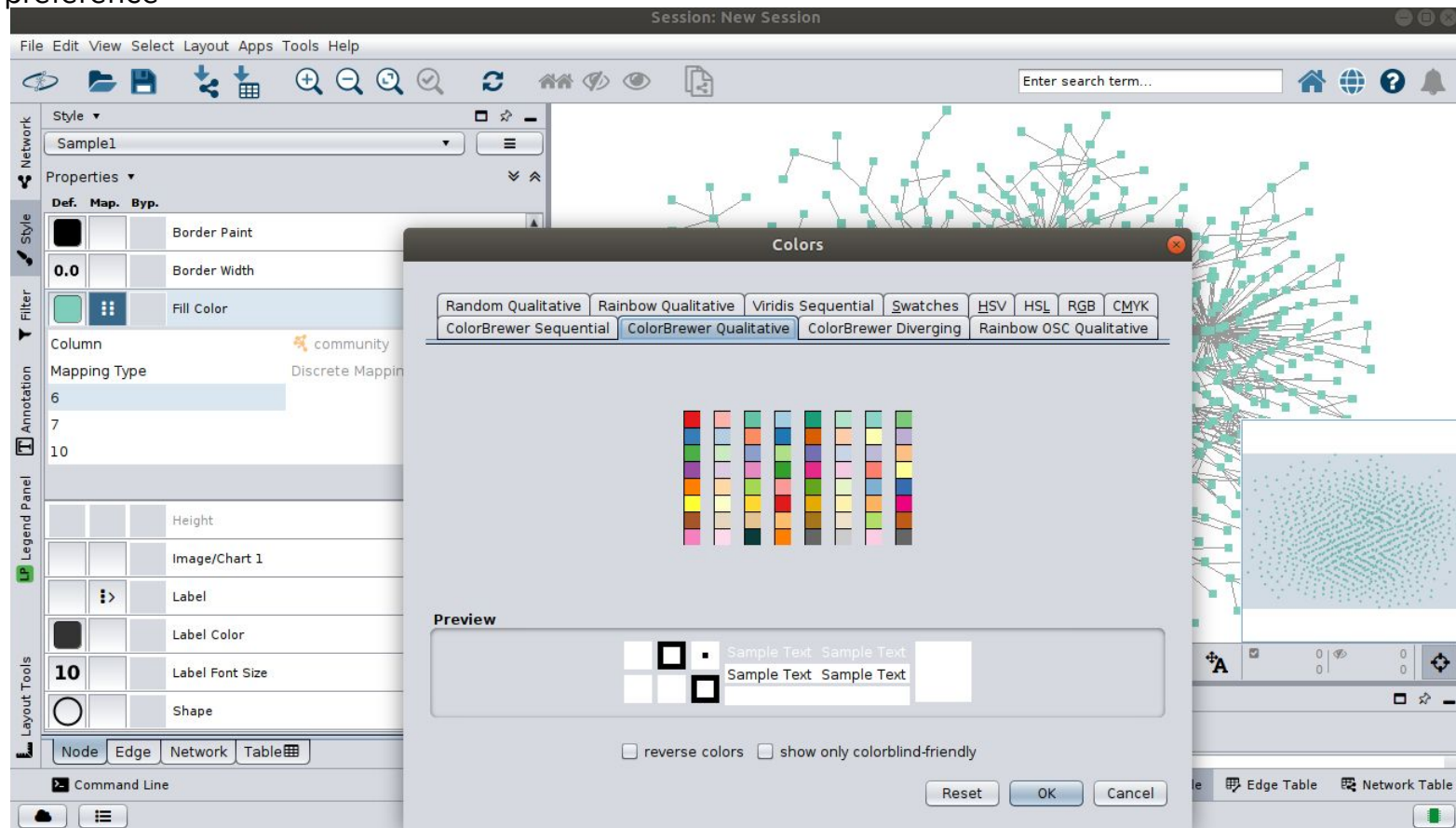
node	community
P25788	10
A0A5B9	10
P01848	10
P30512	10
P78358	10
A0AVI4	10
P55036	10
Q15198	10

Advanced Options...

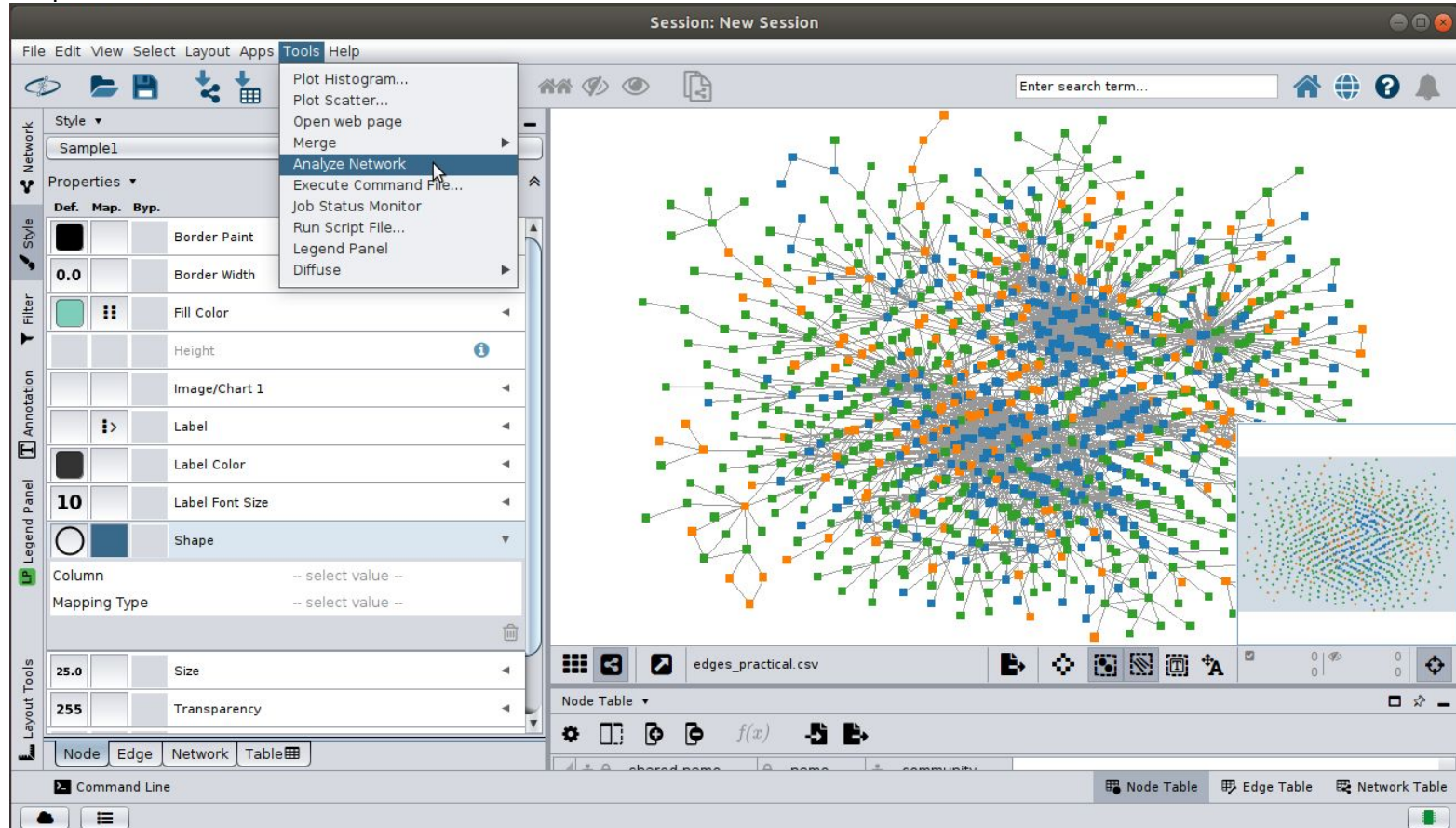
OK Cancel

Node Table Edge Table Network Table

To color the nodes according to community, go to **Fill color** in **Styles**. Select **Column -> community**, **Mapping type -> Discrete** since the communities are only 3. Give unique color to each community according to your preference



To represent the nodes according to various centralities like their degree, betweenness...., go to **Tools** and click on **Analyze Network**. All the properties of the network for each node are computed by the cytoscape software.



Do not select “Directed Graph” since our network is not directed.

The screenshot displays a network visualization software interface. The main window shows a large, complex undirected graph with numerous nodes and edges. The nodes are colored in green, blue, and orange, and are connected by a dense web of edges. A 'Set Parameters' dialog box is open in the center, asking 'Analyze as Directed Graph?' with a checkbox that is currently unchecked. The dialog has 'OK' and 'Cancel' buttons. The interface includes a menu bar (File, Edit, View, Select, Layout, Apps, Tools, Help), a toolbar with various icons, and a sidebar on the left with tabs for Network, Style, Filter, Annotation, Legend Panel, and Layout Tools. The 'Style' tab is active, showing properties for 'Sample1' such as Border Paint, Border Width, Fill Color, Height, Image/Chart 1, Label, Label Color, Label Font Size, Shape, Column, Mapping Type, Size, and Transparency. The bottom status bar shows 'edges_practical.csv' and tabs for Node Table, Edge Table, and Network Table.

Session: New Session

File Edit View Select Layout Apps Tools Help

Enter search term...

Style ▾

Sample1

Properties ▾

Def.	Map.	Byp.	
			Border Paint
0.0			Border Width
			Fill Color
			Height
			Image/Chart 1
			Label
			Label Color
10			Label Font Size
			Shape
Column -- select value --			
Mapping Type -- select value --			
25.0			Size
255			Transparency

Set Parameters

Analyze as Directed Graph? ☐

OK Cancel

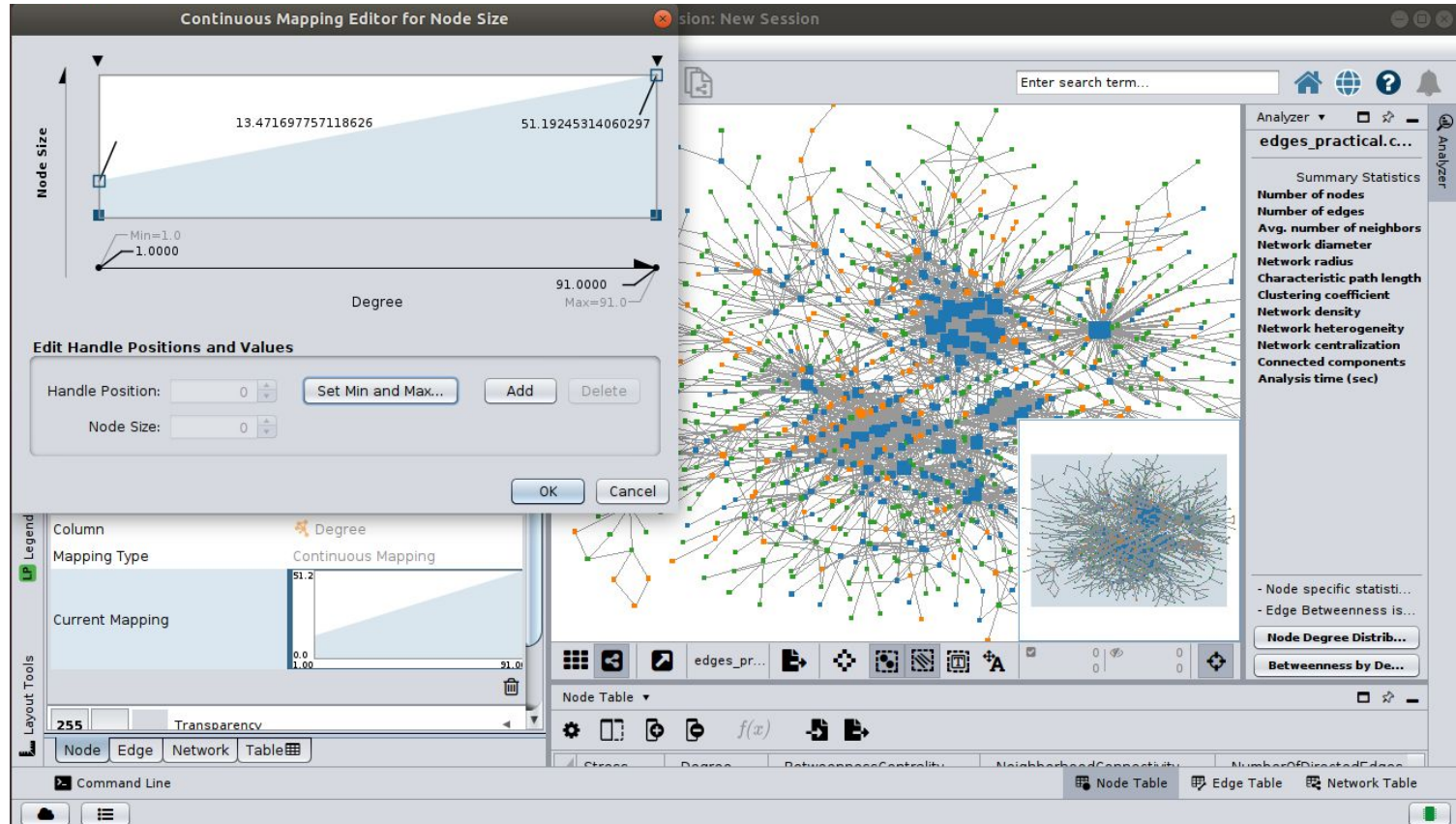
edges_practical.csv

Node Table ▾

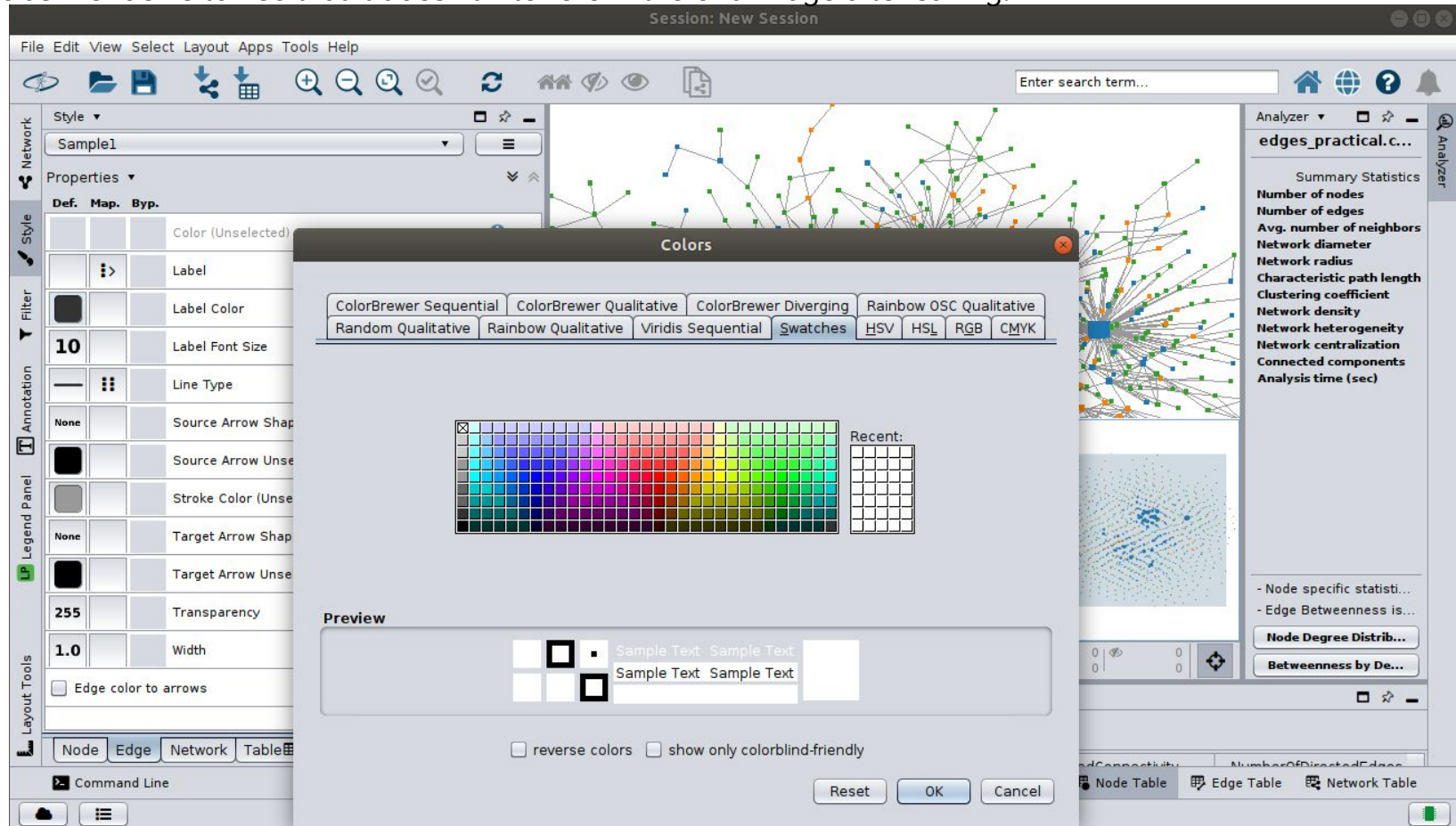
Command Line

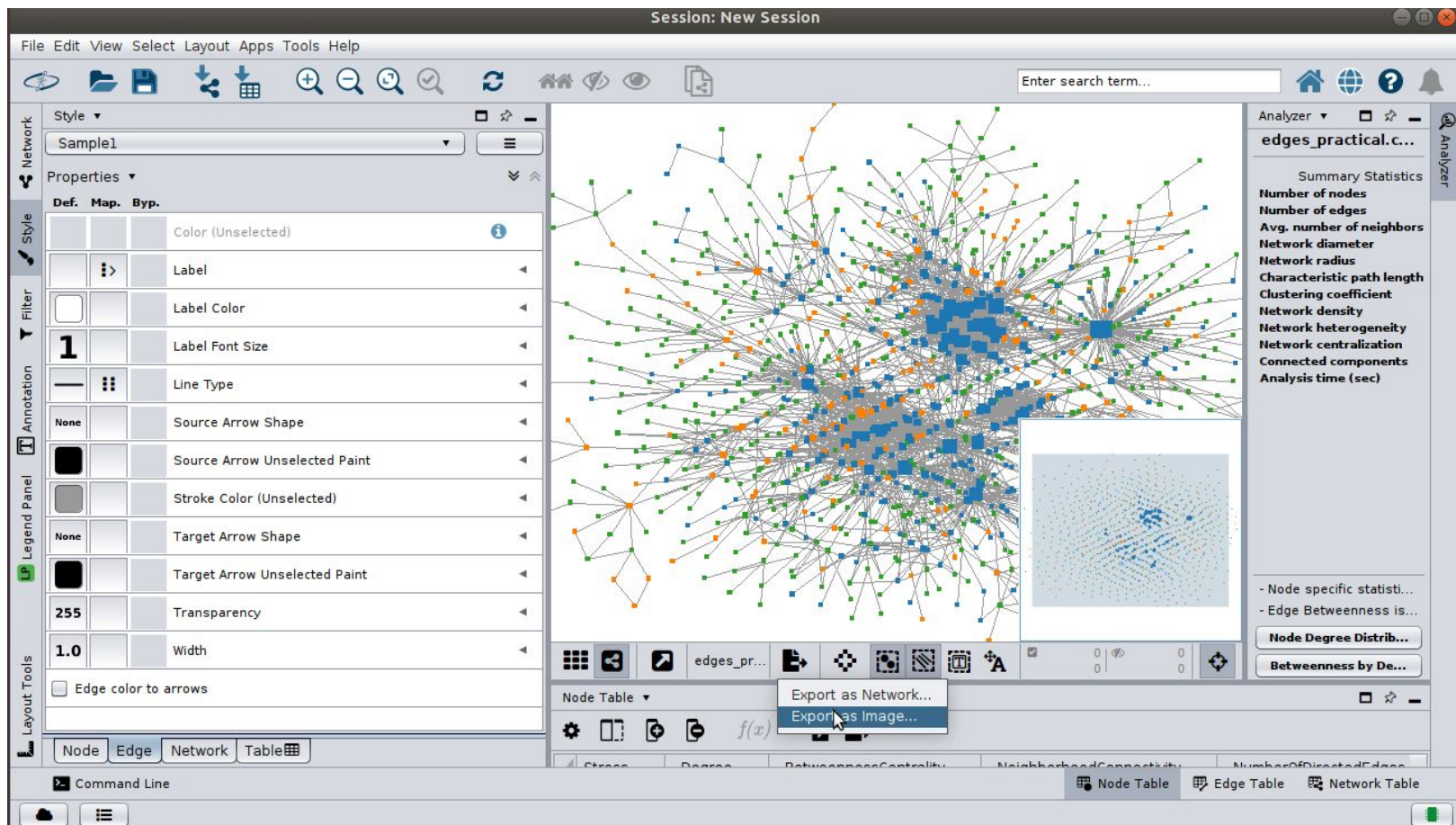
Node Table Edge Table Network Table

To represent the nodes according to degree, go to **Size** in **Styles**. Select **Column** -> **Degree**(which appears only after you “**Analyze network**” as mentioned in previous slides), **Mapping type** -> **Continuous**. Modify the range of the degree in the window to clearly observe that the size of high degree nodes should be bigger relatively to those of low degree nodes.

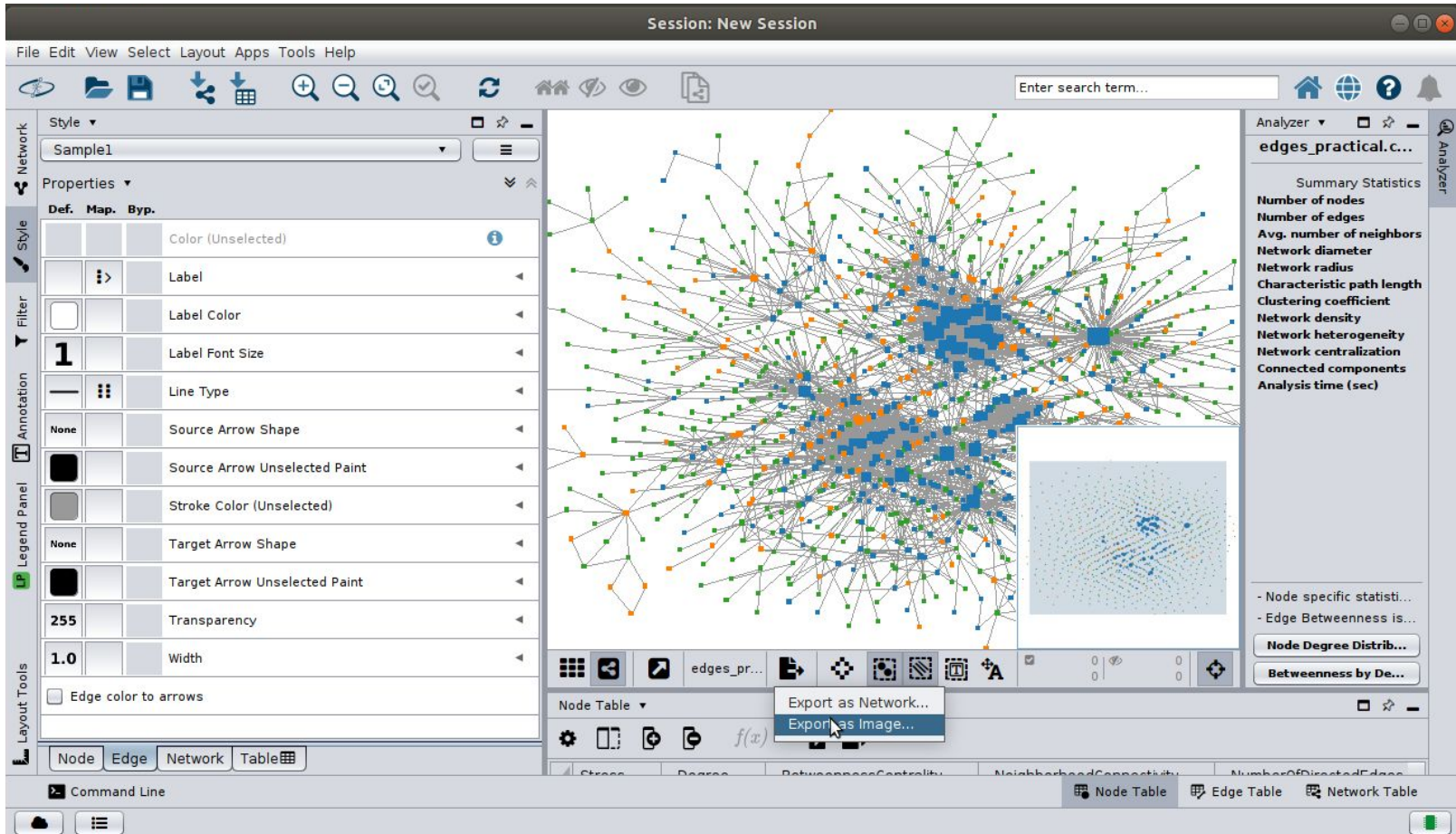


You can possibly try to remove the edge label by setting it to white in the **Label** option of **Edge** and also set **Label Font size** to 1 so that it doesn't interfere in the end image after saving.

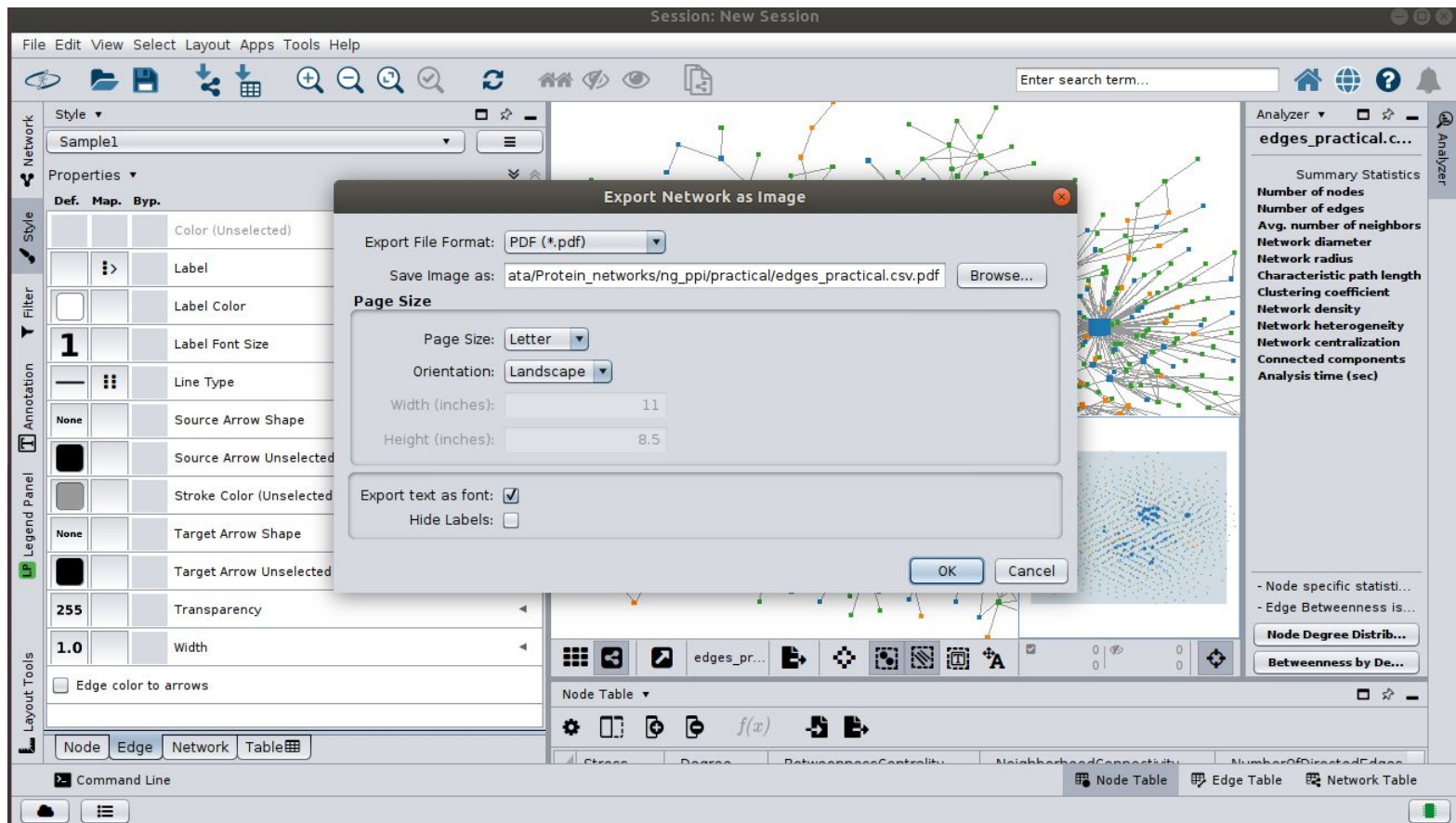




To save your image, go to option shown below and click on **Export as Image**.



Set your preferences as Landscape or Portrait. And also the file format of your image.



To visualize the betweenness centrality of nodes, first you need to remove the existing mapping of the node. Press **Remove Mapping** in the **Fill Color** option for **Node**.

The screenshot displays the Gephi software interface during a 'New Session'. The main window shows a complex network graph with nodes colored in blue, green, and orange, connected by a dense web of edges. A tooltip labeled 'Remove Mapping' is visible over the 'Fill Color' property in the left sidebar.

Left Sidebar (Properties Panel):

- Style:** Sample1
- Properties:** Def. Map. Byp.
- Fill Color:** Discrete Mapping
- Column:** community
- Mapping Type:** 6 (R:255 G:127 B:0 - #FF7F00), 7 (R:51 G:160 B:44 - #33A02C), 10 (R:31 G:120 B:180 - #1F78...)
- Legend Panel:** Height, Image/Chart 1, Label, Label Color, Label Font Size (10), Shape

Right Sidebar (Analyzer Panel):

- edges_practical.c...**
- Summary Statistics:** Number of nodes, Number of edges, Avg. number of neighbors, Network diameter, Network radius, Characteristic path length, Clustering coefficient, Network density, Network heterogeneity, Network centralization, Connected components, Analysis time (sec)
- Node specific statist...**
- Edge Betweenness is...**
- Node Degree Distrib...**
- Betweenness by De...**

Bottom Panel (Node Table):

Stress	Degree	BetweennessCentrality	NeighborhoodConnectivity	NumberOfDirectedEdges

To color the nodes according to its betweenness value, go to **Fill color** in **Styles**. Select **Column** -> **BetweennessCentrality**, **Mapping type** -> **Continuous**. To change color style or the range you can double click on the Current Mapping and modify it to suite your preferences.

